

Topic 4 of Personnel Economics: Non-Classical Motivators

Exercises

Exercise 4.1

A worker chooses effort e . Effort results in high output with probability e and in low output with the remaining probability $1-e$. If the worker achieves high output, the manager pays a bonus b to the worker. In addition, the worker always receives base salary w . The utility function of the worker is total income minus cost of effort $(\theta e^2)/2$. Throughout the exercise you can assume that the participation constraint of the worker is satisfied. The level of θ is such that effort chosen by the worker is always lower than 1.

a) What level of effort does the worker choose?

Suppose now that the worker is loss averse. His marginal utility from income above the reference income is one. His marginal utility from income below the reference income is $R > 1$. The worker's reference income is equal to his base salary w .

b) What level of effort does the worker choose?

Suppose now that the manager offers the following scheme: The worker earns a base salary w plus half of b . If the worker reaches high output, the worker receives a bonus equal to half of b . If the worker reaches low output, the worker must pay back half b . The worker's reference income under the new scheme is w plus half b .

c) What level of effort does the worker choose?

d) Think of a way in which effort chosen by the worker would be even higher.

e) A personnel economist says that contracts that exploit workers' loss aversion may be much less attractive for employers when they operate in a competitive labor market in which they need to make workers at least as well off as when workers take another job. Comment.

Exercise 4.2 *(copied from Peter Kuhn's teaching material)*

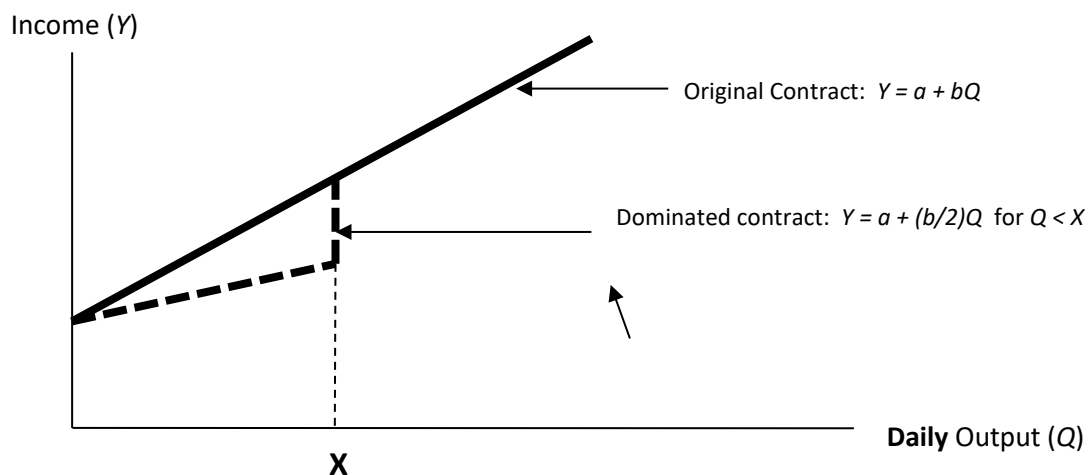
Evaluate the following statement: If a person values utility today more than utility in the future, then his/her preferences exhibit a property economists call present bias.

Exercise 4.3 *(adapted from Peter Kuhn's teaching material)*

a) What is a dominated contract?

b) Why is the contract depicted in the Figure below a dominated contract?

c) Why might sophisticated, present-biased people choose a dominated contract?



Chapter 9, discussion questions 2 and 4. And add to question 4: How can these potential problems be resolved? I.e., how can one get a reliable estimate of the effect of symbolic rewards on performance?

Exercise 4.4

We discussed and played the following gift-exchange game during the main lecture:

“A” gets some money, let’s say 10 euro, from the experimentalist.

A decides how much to transfer to B.

B can return some or all of it to A; the returned amount is doubled by the experimentalist.

End of the game.

Suppose that, instead of doubled, the amount returned by player B is tripled by the experimentalist.

- What share of the amount sent by player A should player B return for player A to be at least as well off financially?
- Do you think it is reasonable to assume that player B returns a fraction α that is independent of the amount sent by player A? If not, do you think it increases or decreases? Discuss why.

Exercise 4.5 (adapted from Peter Kuhn’s teaching material)

Consider two possible utility functions for workers. In one, $U = Y - V(E)$, where $V' > 0$ and $V'' > 0$. In the other, $U = H(Y) - V(E)$, where $H' > 0$, $H'' < 0$, $V' > 0$ and $V'' > 0$.

- Which one of these has diminishing marginal utility of income?
- Which one gives rise to income effects on labor supply?

c) Assume $Q=E$ and $Y = a + bQ$. Use the first-order conditions to explain what effect an increase of the bonus b has on effort E for both utility functions. Do the same for an increase of the base salary a .

Answers

4.1

- a) Maximize $w + eb - (\theta e^2)/2$ to e gives first-order condition $b - \theta e = 0$. Solving gives $e = b / \theta$.
- b) The result is the same as under a). The worker's income will never be in the 'loss range' and so the R plays no role.
- c) Maximize $w + b/2 + e(b/2) - (1-e)R(b/2) - (\theta e^2)/2$ to e gives first-order condition $b/2 + R(b/2) - \theta e = 0$. Solving gives $e = (0.5+0.5R)b / \theta$. Note: if the worker is not loss averse (i.e. $R=1$), the outcome is the same as under a and b. If the worker is loss averse, he works more under this scheme.
- d) Perhaps by paying $w + b$ combined with the obligation to repay b when output is low. If the reference wage in that case would become $w + b$, then the worker chooses effort equal to Rb / θ .
- e) The personnel economist is right. Any effort increase due to such contracts goes hand in hand with an increase in compensation for employers operating in competitive labor markets. Moreover, such contracts can also make that workers sometimes end up in the loss domain, which causes additional utility loss, which need to be compensated by the employer in order to remain an attractive employer. Note that these effects are not present when the worker earns a rent, e.g. due to a high minimum wage or due to collective wage bargaining.

4.2 *(copied from Peter Kuhn's teaching material)*

False. Even perfectly "rational" classic economic agents may value utility today more than utility in the future i.e. their discount factor, δ , can be less than one. This is because all sorts of uncertainty exists, and our lives are finite in any case: you are less likely to be alive at dates further in the future than at more recent dates so it make sense to place less emphasis on utility at later dates.

Present bias ($\beta < 1$) refers to an additional type of discounting that places a lower value on any consumption that does not occur right now in the present. The key difference this causes is that present-biased individuals will change their plans simply because today has arrived, despite not learning any new information that should change their plans. In contrast, traditional rational economic agents will make time-consistent decisions, and will not change their behavior unless new information comes up.

4.3

- a) Contract A dominates contract B if contract A for any realization of output or the state of nature is at least as good and sometimes better than contract B.
- b) The original contract is strictly better up to $Q=X$ and identical for $Q>X$. Hence, the original contract dominates the other contract.
- c) It may work as a commitment device. With the original contract, a present-biased worker chooses too low effort from the perspective of the present self. The dominated contract introduces a penalty for slacking off, inducing the worker to work hard, which is in the interest of present self who chooses the contract.

Answers to Chapter 9, discussion questions 2 and 4:

<https://drive.google.com/file/d/1EJ9D5a0Z8l1TTpRValUbS7XPakRrD4rJ/view>

4.4

- a) Player A is at least as well off when $10 - X + \alpha * 3 * X \geq 10$, and so when $\alpha \geq (1/3)$.
- b) This is, in the end, an empirical question. Several theories predict that the amount returned increases with the amount sent (fairness concerns, intention-based reciprocity). Whether this is a linear relation depends on details of the theoretical model.

4.5

- a) The utility function $U = Y - V(E)$, where $V' > 0$ and $V'' > 0$ has constant marginal utility of income (because utility is linear in income.) The utility function $U = H(Y) - V(E)$, where $H' > 0$, $H'' < 0$, $V' > 0$ and $V'' > 0$ has diminishing marginal utility of income because the marginal utility of income, H' , is diminishing with income.
- b) Income effects will arise when we use a utility function with diminishing marginal utility of income. In contrast, $U = Y - V(E)$ is a utility function that has no income effects.
- c) The first-order condition with the utility function $U = Y - V(E)$ is:

$$b - V'(E) = 0$$

Hence, base salary a has no influence on E . And the bonus b increases E , because the marginal benefit from effort increase with the bonus.

The first-order condition with the utility function $U = H(Y) - V(E)$ is:

$$bH'(a + bE) - V'(E) = 0$$

Hence, a decreases E , because a higher base salary means that the marginal utility from income is lower and hence working more has less of an effect on utility from income. The effect of the bonus becomes indeterminate: A higher bonus increases the marginal benefit from effort through the first b (outside the function $H'()$), which means that each unit of effort results in a higher increase in income when the bonus is higher, and hence more effort is optimal. On the other hand, for a given level of effort, an increase in the bonus increases income, and so decreased $H'()$, and so makes working less an optimal response. The sum of these two effects can be positive or negative. Hence, we can have a “backward bending effort curve”.